Quantification: 699_HYP_HP_z20c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1438.44	1239	799.765	px^2
major_axis_length_px	lacuna shape	70.1037	63.7	22.8233	px
minor_axis_length_px	lacuna shape	33.4837	30.7419	15.9453	px
aspect_ratio	lacuna shape	2.5837	1.665	1.5667	unitless
eccentricity	lacuna shape	0.8208	0.7995	0.1366	unitless
solidity	lacuna shape	0.7565	0.7599	0.117	unitless
roots_count	headline	10.25	7.5	13.4714	count
ring_length_r30_px	headline	368.5	288	404.494	px
ring_length_r60_px	reported	697.083	650.5	643.89	px
owned_length_px	ownership-dependent	748.025	609.138	894.531	px
edge_count	ownership-dependent	42.75	36.5	53.5285	count
mean_edge_length_px	ownership-dependent	17.615	18.0343	2.5023	px
perimeter_px	normalised	323.502	189.965	362.545	px
ring_area_r30_px2	normalised	9610.67	7018	6578.44	px^2
ring_area_r60_px2	normalised	21960.1	17966.5	11085.4	px^2
in_frame_fraction_r30	normalised	0.9802	1	0.0685	unitless
in_frame_fraction_r60	normalised	0.9722	1	0.0964	unitless
ring_density_r30	normalised	0.0346993	0.0319007	0.0177191	px^-1
ring_density_r60	normalised	0.0293536	0.0341583	0.0158372	px^-1
roots_per_100px_perimeter	normalised	3.0251	3.0135	1.609	per 100 px
ring_attached_length_r30_px	network v2	323.5	209.5	402.358	px
ring_attached_length_r60_px	network v2	539.25	342	638.532	px
ring_length_w_r30_px	network v2	411.676	317.587	456.413	px
ring_length_w_r60_px	network v2	782.487	726.514	732.824	px
sholl_crossings_r10	network v2	8.0833	5.5	9.9312	count
sholl_crossings_r20	network v2	8.75	7	7.7357	count
sholl_crossings_r30	network v2	10.0833	9.5	8.7226	count
loop_count	loops, report only	1.6667	1	2.8069	count
loop_edge_share	loops, report only	0.0908	0.0641	0.1024	share
ring_width_mean_r30_px	width	6.6979	6.6508	1.1383	px

Per field
----

measure	value	units
analysed_area_px2	877006	px^2
total_skeleton_length_px	16062	px
canalicular_length_density_per_px	0.0183146	px^-1
junction_count	301	count
junction_density_per_px2	0.00034321	px^-2
lacuna_count	12	count

lacunae_per_px2	1.368e-05	px^-2
skeleton_component_count	313	count
mean_component_length_px	51.3163	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0205096	px^-1
field_density_without_flagged_per_px	0.0141702	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	964	px
unattached_component_count	48	count
canalicular_length_density_attached_per_px	0.0183146	px^-1
field_length_density_with_unattached_per_px	0.0194138	px^-1
excluded_area_px2	147318	px^2
analysed_field_area_px2	901258	px^2
loop_count	24	count
loop_edge_share	0.0885	share
edge_count	1017	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.9508	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	13904	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
2	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
3	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
4	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
5	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
6	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
7	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
8	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
9	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False
10								283.342	208.259	False
11								341.9	874.582	False
12								546.189	775.469	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	241	741	991.553	51	19.4422	8.4329
2	8	343	755	809.838	50	16.1968	6.2878
3	9	401	809	774.713	44	17.6071	7.0585
4	6	192	446	345.238	17	20.3081	6.6589
5	4	335	560	443.563	29	15.2953	7.4557

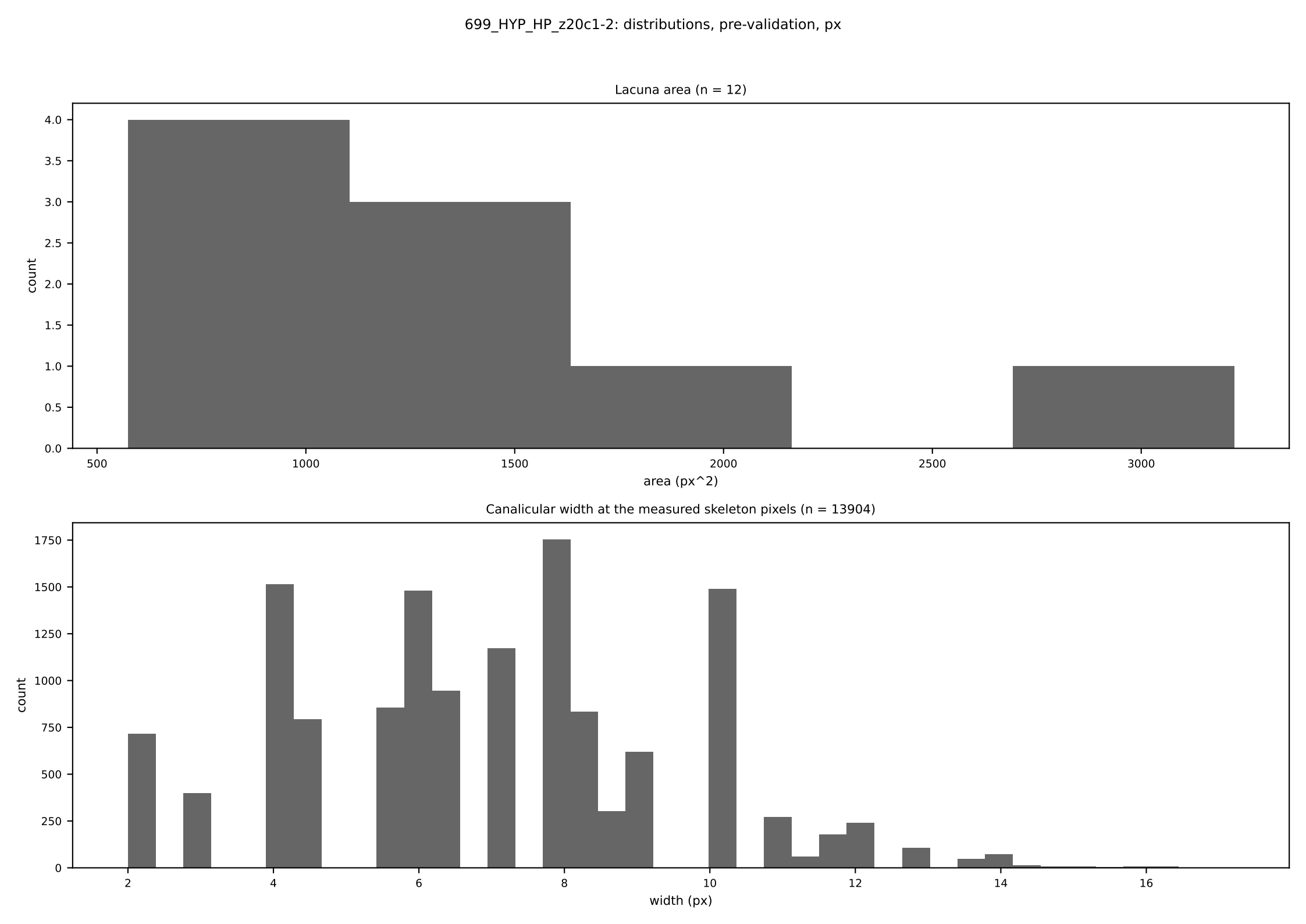
6	16	545	972	1163.08	63	18.4615	7.842
7	7	348	808	1017.65	49	20.7683	6.0891
8	9	161	333	127.669	10	12.7669	6.6426
9	2	35	62	18.4853	1	18.4853	5.146
10	51	1570	2515	3248.96	197	16.4922	8.2003
11	1	94	153	20.4853	1	20.4853	5.3077
12	2	157	211	15.0711	1	15.0711	5.2531

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	185.581	7275	19284	1	1	0.0331272	0.0384256	4.3108
2	418.517	11182	24664	1	1	0.0306743	0.0306114	1.9115
3	221.179	7585	17507	1	1	0.0528675	0.0462101	4.0691
4	140.326	6761	16630	1	1	0.0283982	0.026819	4.2758
5	165.74	6360	13932	1	1	0.052673	0.0401952	2.4134
6	316.149	10460	25779	1	1	0.0521032	0.0377051	5.0609
7	132.267	6479	17177	1	1	0.053712	0.0470397	5.2923
8	363.747	11289	26807	1	1	0.0142617	0.0124221	2.4742
9	163.681	5591	13507	0.7628	0.6662	0.00626006	0.00459021	1.2219
10	1435.5	29498	54199	1	1	0.0532239	0.0464031	3.5528
11	144.995	6436	18426	1	1	0.0146053	0.00830348	0.6897
12	194.35	6412	15609	1	1	0.0244853	0.0135178	1.0291

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	203	646	268.966	827.784	4	6	10
2	216	332	386.433	843.11	9	9	12
3	379	668	448.363	905.997	7	10	9
4	184	253	212.297	498.818	3	6	4
5	255	352	366.208	625.245	7	7	10
6	543	767	624.014	1112.56	15	13	12
7	298	684	387.25	905.169	6	12	13
8	134	204	175.125	362.25	5	2	6
9	24	24	40.4558	70.8406	0	0	1
10	1515	2388	1764.63	2850.97	37	30	35
11	54	67	99.8701	163.397	1	3	5
12	77	86	166.498	223.711	3	7	4


699_HYP_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 13904)
count
width (px)